

3D visualization of multiway networks

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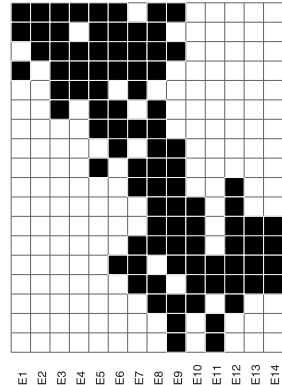
Outline

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<https://github.com/bavla/ibm3m/>

European airports and flight companies dataset

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The *European airports and flight companies* dataset was collected by A. Cardillo and his collaborators for the paper

A. Cardillo, J. Gómez-Gardeñes, M. Zanin, M. Romance, D. Papo, F. del Pozo, S. Boccaletti: *Emergence of Network Features from Multiplexity*, *Scientific Reports* 3, 1344 (2013).

It is an example of a multiway network. It contains 450 airports making two ways *airA* (departure airports) and *airB* (arrival airports), the third way is *line* (37 flight companies). There are 7176 links $(airA, airB, line) \in \mathcal{L}$ telling that there is a flight provided **by** the company *line* **from** the airport *airA* **to** the airport *airB*. Each link has the weight $w = 1$.

The multiway network AirEu2013 represented in JSON format and an R package MWnets for analysis of multiway networks are available at [GitHub/Bavla/ibm3m](https://github.com/Bavla/ibm3m).

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Raw data

e	airA	airB	line	w
1	LCLK	EDDF	Lufthansa	1
2	LCLK	EDDM	Lufthansa	1
3	EDDF	LCLK	Lufthansa	1
4	EDDF	LFPG	Lufthansa	1
5	EDDF	LFBO	Lufthansa	1
6	EDDF	EDDP	Lufthansa	1

Factors

e	airA	airB	line	w
1	1	2	1	1
2	1	38	1	1
3	2	1	1	1
4	2	7	1	1
5	2	8	1	1
6	2	10	1	1

7171	LGSK	LGAV	Olympic Air	1
7172	LGKC	LGAV	Olympic Air	1
7173	LGIO	LGAV	Olympic Air	1
7174	LGPL	LGAV	Olympic Air	1
7175	LGK	LGAV	Olympic Air	1
7176	LGX	LGAV	Olympic Air	1

7171	354	14	37	1
7172	393	14	37	1
7173	404	14	37	1
7174	426	14	37	1
7175	428	14	37	1
7176	444	14	37	1

LCLK – Larnaca International Airport; EDDF – Frankfurt Airport; LGAV – Athens International Airport (Eleftherios Venizelos Airport)

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A *weighted multiway network* $\mathcal{N} = (\mathcal{V}, \mathcal{L}, w)$ [3] is based on *nodes* from k finite sets (ways or dimensions) $\mathcal{V} = (\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_k)$, the set of *links* $\mathcal{L}$, and the *weight* $w : \mathcal{L} \rightarrow \mathbb{R}$. The incidence function $I : \mathcal{L} \rightarrow \mathcal{V}_1 \times \mathcal{V}_2 \times \dots \times \mathcal{V}_k$ assigns to each link $e \in \mathcal{L}$ a k -tuple of its nodes $I(e) = (e(1), e(2), \dots, e(i), \dots, e(k))$, $e(i) \in \mathcal{V}_i$. If for $i \neq j$, $\mathcal{V}_i = \mathcal{V}_j$, we say that $\mathcal{V}_i$ and $\mathcal{V}_j$ are of the same *mode*.

In a general multiway network, different additional data can be known for nodes and/or links $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$, where $\mathcal{P}$ is a set of node properties $p : \mathcal{V}_i \rightarrow S_p$, and $\mathcal{W}$ is a set of link weights $w : \mathcal{L} \rightarrow S_w$.

Some 3-way datasets [4].

Simple networks and subnetworks

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A multiway network $\mathcal{N}$ is *simple* if and only if all its link tuples are mutually different – for all $e, f \in \mathcal{L}$ it holds $e \neq f \Rightarrow I(e) \neq I(f)$.

A multiway network $\mathcal{N}' = (\mathcal{V}', \mathcal{L}', \mathcal{P}', \mathcal{W}')$ is a *subnetwork* of the multiway network $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$ if and only if it can be obtained from $\mathcal{N}$ by removing:

- 1 some nodes; if a node $v \in \mathcal{V}_i$ is removed then also all links $e \in \mathcal{L}$ that contain it, $e(i) = v$, are removed.
- 2 some links.
- 3 some ways.
- 4 some properties from $\mathcal{P}$ or $\mathcal{W}$.

Multiway network representation in R

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We decided to use for multiway networks the following representation in R

```
MW <- list(  
  format="MWnets",  
  info= metadata,  
  ways= list of ways' names,  
  nodes= list of ways' dataframes,  
  links= dataframe of links and weights,  
  data= list of additional data  
)
```

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```
List of 6
$ format: chr "MWnets"
$ info :List of 7
..$ network: chr "AirEu2013"
..$ title : chr "Air Transportation Multiplex"
..$ ref : chr "Cardillo A. et al. Emergence of network features from multiplexity, Scientific
..$ href : chr "http://complex.unizar.es/~atnmultiplex/"
$ ways :List of 3
..$ airA: chr "first airport"
..$ airB: chr "second airport"
..$ line: chr "airline"
$ nodes :List of 3
..$ airA:'data.frame': 450 obs. of 7 variables:
.. ..$ ID : chr [1:450] "LCLK" "EDDF" "EDDK" "EGNX" ...
.. ..$ lon : num [1:450] 33.63 8.57 7.14 -1.33 -3.41 ...
.. ..$ lat : num [1:450] 34.9 50 50.9 52.8 50.7 ...
.. ..$ iata : chr [1:450] "LCA" "FRA" "CGN" "EMA" ...
.. ..$ long : chr [1:450] "Larnaca International Airport" "Frankfurt Airport" "Cologne Bonn
.. ..$ region : chr [1:450] "Larnaka" "Hessen" "Nordrhein-Westfalen" "England" ...
.. ..$ country: chr [1:450] "CY" "DE" "DE" "GB" ...
..$ airB:'data.frame': 450 obs. of 2 variables:
.. ..$ ID : chr [1:450] "LCLK" "EDDF" "EDDK" "EGNX" ...
.. ..$ long: chr [1:450] "Larnaca International Airport" "Frankfurt Airport" "Cologne Bonn Air
..$ line:'data.frame': 37 obs. of 1 variable:
.. ..$ ID: chr [1:37] "Lufthansa" "Ryanair" "Easyjet" "British A" ...
$ links :'data.frame': 7176 obs. of 4 variables:
..$ airA: int [1:7176] 1 1 2 2 2 2 2 2 2 ...
..$ airB: int [1:7176] 2 38 1 7 8 10 14 15 17 18 ...
..$ line: int [1:7176] 1 1 1 1 1 1 1 1 1 ...
..$ w : int [1:7176] 1 1 1 1 1 1 1 1 1 ...
$ data :List of 1
..$ Eu:'data.frame': 40 obs. of 6 variables:
.. ..$ alpha2 : chr [1:40] "AT" "BA" "BE" "BG" ...
.. ..$ alpha3 : chr [1:40] "AUT" "BIH" "BEL" "BGR" ...
.. ..$ Ccode : int [1:40] 40 70 56 100 756 196 203 276 208 233 ...
.. ..$ long : chr [1:40] "Austria" "Bosnia and Herzegovina" "Belgium" "Bulgaria" ...
.. ..$ region : chr [1:40] "Europe" "Europe" "Europe" "Europe" ...
.. ..$ subregion: chr [1:40] "Western Europe" "Southern Europe" "Western Europe" "Eastern Europe"
```

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3D visualization of multiway networks

For the analysis of multiway networks, we are developing an R package **MWnets**. It supports transformations of multiway networks such as reordering ways, slicing, extracting subnetwork, flattening, reordering links, joining ways, Salton's cosine, computing cores, etc.

For a visualization of 3-way slices a generalization of matrix representation of networks to 3D is used. It is supported by the function `mnwX3D` that creates a corresponding layout description in X3D format. It can be inspected using special viewers (**view3dscene**) or in a web browser using the Javascript library **X3DOM**.

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A list of subways $\mathcal{C} = (C_1, C_2, \dots, C_r)$, $C_i \subseteq \mathcal{V}_{h(i)}$ we will call a *selection*. Note that the selection $\mathcal{C}$ defines a function $h : 1..r \rightarrow 1..k$. For a link $e \in \mathcal{L}$ we introduce abbreviations

$$h(e) \equiv (e(h(1)), e(h(2)), \dots, e(h(r)))$$

for a *selected part* of the link e ; and

$$h(e) \in \mathcal{C} \equiv \forall i \in 1..r : e(h(i)) \in C_i$$

for a statement that all nodes of the selected part of the link e belong to selected subways.

For a node $u \in \mathcal{V}_i$ and a selection $\mathcal{C}$, we call a *star* in the node u for the selection $\mathcal{C}$, the set

$$S(u, \mathcal{C}) = \{e \in \mathcal{L} : e(i) = u \wedge h(e) \in \mathcal{C}\}.$$

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For a node $u \in \mathcal{V}_i$ we call the set of *neighbors* of the node u in the way $\mathcal{V}_j$ for the selection $\mathcal{C}$, the set of nodes $N(u, \mathcal{C}, \mathcal{V}_j) = \mathcal{V}_j(S(u, \mathcal{C}))$.

Using stars and sets of neighbors we can define some node property functions on multiway networks. For a node $u \in \mathcal{V}_i$ and a selection $\mathcal{C}$

degree: $p_d(u, \mathcal{C}) = \text{card}(S(u, \mathcal{C}))$

weighted degree for the weight w : $p_s(u, \mathcal{C}; w) = \sum_{e \in S(u, \mathcal{C})} w(e)$

diversity of the way $\mathcal{V}_j$: $p_\delta(u, \mathcal{C}, \mathcal{V}_j) = \text{card}(N(u, \mathcal{C}, \mathcal{V}_j))$

We say [2] that the node property function $p(v, \mathcal{C})$:

- is *local* iff: $p(v, \mathcal{C}) = p(v, N(v, \mathcal{C}))$ for all $v \in \mathcal{V}$.
- is *monotonic* iff: $\mathcal{C}_1 \subset \mathcal{C}_2 \Rightarrow \forall v \in \mathcal{V} : p(v, \mathcal{C}_1) \leq p(v, \mathcal{C}_2)$.

Cores in multiway networks

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Let $\mathcal{C} = (C_1, C_2, \dots, C_k) \subseteq \mathcal{V}$ be a selection, $\mathbf{p} = (p_i)$ be a list of monotonic node property functions over the selected ways, and $\mathbf{t} = (t_i)$ a list of the corresponding thresholds.

The multiway subnetwork $\text{Core}(\mathbf{p}, \mathbf{t}) = (\mathcal{C}, \mathcal{L}(\mathcal{C}))$ induced by the selection $\mathcal{C}$ in a multiway network $\mathcal{N}$ is a *generalized multiway core* for node property functions $\mathbf{p}$ at levels $\mathbf{t}$ if and only if it holds that for all $v \in C_i : p_i(v, \mathcal{C}) \geq t_i$ and $\mathcal{C}$ is the maximal such selection.

Then for monotonic node property functions $\mathbf{p}$ the generalized core can be determined by the following algorithm

- 1: **function** COREMW($\mathcal{N}, \mathbf{p}, \mathbf{t}$)
- 2: $\mathcal{C} \leftarrow (\mathcal{V}_{h(1)}, \mathcal{V}_{h(2)}, \dots, \mathcal{V}_{h(r)})$
- 3: **while** $\exists u \in C_i : p_i(u, \mathcal{C}) < t_i$ **do** $C_i \leftarrow C_i \setminus \{u\}$
- 4: **return** $\mathcal{C}$
- 5: **end function**

Screenshot of a part of the 3D layout of AirEu2013

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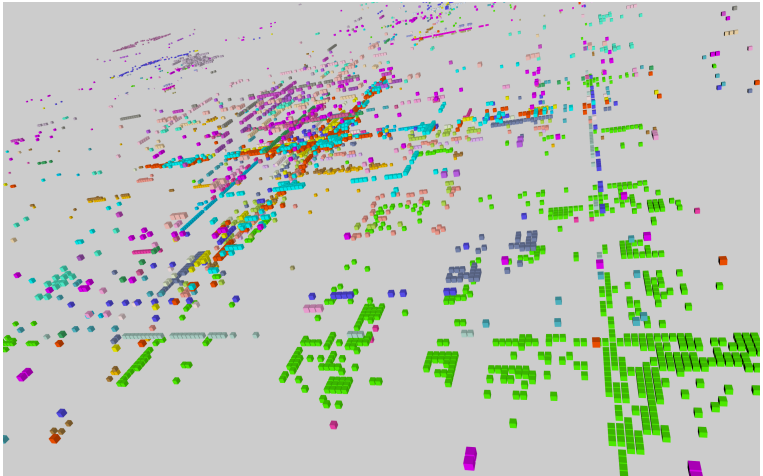
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3D layout: The green links in the bottom right corner are provided by Ryan air – it is mostly linking airports that are not linked by other companies.

3D layout of EUair diversity core at level 13

initial (left) and reordered by crosses (right)

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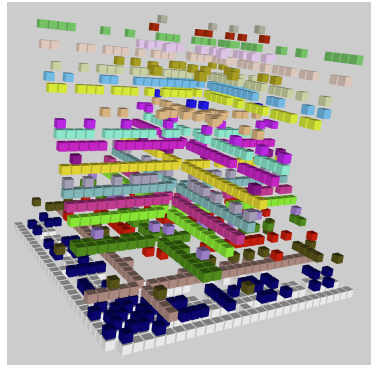
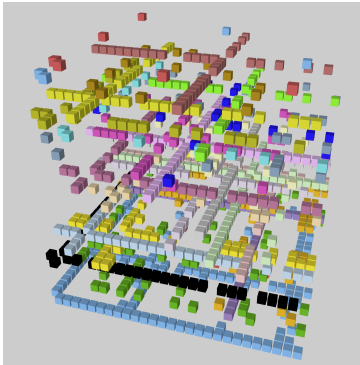
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(1) Frankfurt Ap, (2) Berlin Tegel Ap, (3) Munich Ap, (4) Dusseldorf Ap, (5) Hamburg Ap, (6) Zurich Ap, (7) Geneva Ap, (8) Milan-Malpensa Ap, (9) Copenhagen Ap, (10) Stockholm Arlanda Ap, (11) Heathrow Ap, (12) Warsaw Chopin Ap, (13) Vienna IAp, (14) Amsterdam Ap Schiphol, (15) Charles de Gaulle Ap (Roissy Ap), (16) Adolfo Suarez Madrid-Barajas Ap, (17) Barcelona El Prat Ap, (18) Malaga Ap, (19) Vaclav Havel Ap Prague, (20) Leonardo da Vinci-Fiumicino Ap, (21) Brussels Ap (Zaventem Ap), (22) Budapest Ferenc Liszt IAp, (23) Athens IAp (Eleftherios Venizelos Ap), (24) Ben Gurion Ap, (25) Henri Coanda IAp, (26) Venice Marco Polo Ap, (27) Sofia Ap, (28) Nice Cote d'Azur Ap.

Initial Core 3D layout; Reordered Core 3D layout

3D layout of EUair diversity core at level 13

two views exposing companies based on multiple airports

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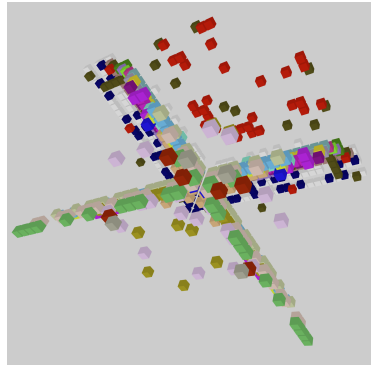
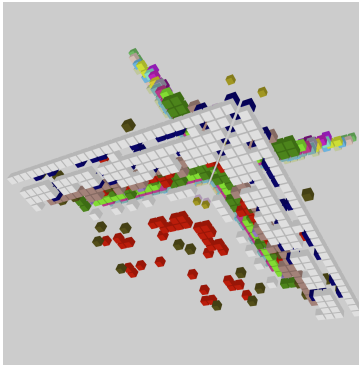
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(1) Lufthansa, (2) Air Berlin, (3) Swiss IAL, (4) Netjets, (5) Easyjet, (6) SAS, (7) Norwegian AS, (8) British A, (9) LOT Polish A, (10) Austrian A, (11) Niki, (12) KLM, (13) Transavia H, (14) Air France, (15) Iberia, (16) Air Nostrum, (17) Vueling A, (18) Ryanair, (19) Czech A, (20) Alitalia, (21) Brussels A, (22) European AT, (23) Malev HA, (24) Wizz Air, (25) Aegean A (26) Olympic Air, (27) TNT Airways.

The companies that were not providing any line between core airports are Turkish A, Flybe, TAP Portugal, Finnair, Air Lingus, Germanwings, Pegasus A, SunExpress, Air Baltic, and Wideroe.

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- Lufthansa and Air Berlin are serving lines from German airports Frankfurt, Berlin, Munich, Dusseldorf, Hamburg, and Milan in Italy
- Swiss IAL and Netjets are serving lines from Swiss airports Zurich and Geneva
- SAS and Norwegian AS are serving lines from Copenhagen and Stockholm
- Spanish airports Madrid, Barcelona, and Malaga are served by companies Iberia, Air Nostrum, Vueling, and Ryanair
- companies serving lines from their base airport: (British A, Heathrow), (LOT, Warsaw), (Austrian A, Niki; Vienna), (KLM, Transavia; Amsterdam), (Air France, CDG), (Czech A, Prague), (Alitalia, Fiumicino), (Brussels A, European AT; Brussels), (Malev, Budapest), (Aegean, Olympic Air; Athens); the only “irregular” link is by Iberia between Barcelona and Budapest

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- Companies with dispersed services inside the core are Easyjet, Netjets, Wizz Air, European AT, and TNT Airways
 - Easyjet is serving lines from Milan and also from CDG, Amsterdam, Fiumicino, Madrid
 - Netjets is serving lines also from Barcelona, Venice, Vienna, and Nice
 - Wizz Air is serving lines from Budapest and Fiumicino
 - European AT has also some lines from Heathrow, Milan, Venice, Barcelona, and Madrid
 - TNT Airways is linking airports CDG and Athens, and Sofia and Henri Coanda

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The data set *120 years of Olympic history – athletes and results* with basic bio data on athletes and medal results from Athens 1896 to Rio 2016 is available at Kaggle [5]. The original data describe 134732 participants of the Summer and Winter Olympics. We transformed it into the multiway network Olympics16S about the medalists of the Summer Olympics till 2016 (available at GitHub/Bavla [1]). It contains data about 34088 medals. Omitting the athlete Name info and simplifying the network we obtain a multiway network with 10429 links on five ways (Games (29), NOC (147), Sport (52), Sex (2), Medal (3)), the weight w counting the medals, and additional weights Age, Height, and Weight.

We decided to search for a core with a large number of medals on the ways Games, NOC, and Sport.

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After examining the distributions of values of the node property function p_S on all three ways we decided to use the core conditions list P that contains three triples

$(p = pWsum, t = 100, args = (Games, NOC, Sport))$,
 $(p = pWsum, t = 100, args = (NOC, Games, Sport))$, and
 $(p = pWsum, t = 100, args = (Sport, Games, NOC))$.

The obtained core was too large. We increased the thresholds to $(500, 300, 350)$. The corresponding p_S -core $\mathcal{C} = (C_{Games}, C_{NOC}, C_{Sport})$ is of order $(577, 308, 390)$ – inside the core, each Olympics accounts for at least 577 medals, each country (NOC) for at least 308 medals, and each sport discipline for at least 390 medals.

Core

... Olympics core

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C_{Games} (25): 1988 Seoul (1405), 1984 Los Angeles (1354), 2008 Beijing (1352), 2012 London (1338), 2000 Sydney (1324), 2016 Rio de Janeiro (1320), 2004 Athens (1305), 1980 Moscow (1270), 1996 Atlanta (1251), 1976 Montreal (1220), 1992 Barcelona (1112), 1972 Munich (1080), 1920 Antwerp (1054), 1964 Tokyo (952), 1968 Mexico City (949), 1912 Stockholm (853), 1960 Rome (826), 1952 Helsinki (797), 1956 Melbourne + Stockholm (795), 1936 Berlin (754), 1948 London (731), 1924 Paris (681), 1908 London (656), 1928 Amsterdam (610), 1932 Los Angeles (577).

C_{NOC} (29): USA (4153), URS (2027), GBR (1594), GER (1573), ITA (1325), FRA (1184), AUS (1165), HUN (1052), SWE (1019), NED (854), GDR (843), RUS (776), CHN (662), JPN (661), ROU (632), CAN (613), NOR (546), POL (520), DEN (519), FRG (500), FIN (446), BRA (446), ESP (416), YUG (379), SUI (352), KOR (347), BEL (337), BUL (317), TCH (308).

C_{Sport} (21): Athletics (2911), Swimming (2815), Rowing (2535), Gymnastics (2043), Fencing (1584), Football (1216), Cycling (1129), Hockey (1126), Canoeing (1010), Shooting (995), Sailing (995), Wrestling (941), Handball (924), Basketball (914), Equestrianism (883), Water Polo (876), Volleyball (839), Boxing (596), Judo (425), Weightlifting (419), Diving (390).

Screenshot of a 3D layout of Olympics weighted degree core at levels (577, 308, 390)

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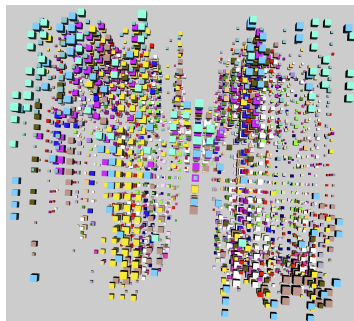
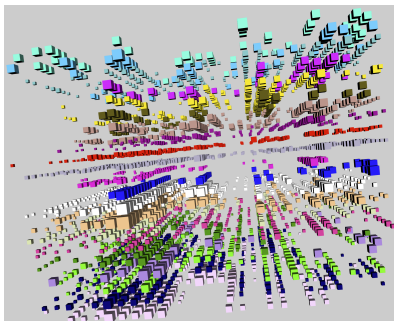
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It turns out that the core contains $25566/34088 = 75\%$ of all medals, $3607/10429 = 34.6\%$ of all links, and occupies 6.9% of the total space. Note also that the set C_{NOC} contains the Soviet Union and Russia; and Germany, West Germany, and East Germany.

The links belonging to the same sport discipline are of the same color. The Olympics are ordered chronologically. The order of NOCs (countries) and sport disciplines was determined using clustering.

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- the R package `MWnets` is still in development (additional methods, objects?, `dplyr`?, robustness)
- additional examples of multiway networks are needed (testing, new problems)
- Python and/or Julia version

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We would like to thank Andreas Plesch for extending the library X3DOM with support for Anchor's attribute description.

The computational work reported in this presentation was performed using R library MWnets. The code and data are available at Github/Bavla [1].

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<https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>